1. Problem and user

- **User:** clinic coordinators in partner counties
- **Problem:** urgent patient messages sit in one queue with routine delivery and catalog questions
- **Decision supported:** which message needs action first, and where supplies are
- **Success:** golden eval stays at 1.00, the fixture agrees with the router, and an urgent message returns `urgency: high`

2. Path chosen and why

- **Path:** AfyaPlus Module 4 Capstone
- **Why:** it combines every module skill (API, auth, RAG, MCP tool, tests, CI) in one product
- **Scope:** stub mode with synthetic data, so it runs offline with no live key
- **Out of scope:** training a model, live crawling of PII, mixing clinical and marketplace fixtures

3. Architecture

```
Client -> /token (JWT login) -> /triage (coordinator only)
                                      |
                    router (config.json keywords)
                   /                  |                  \
          triage model          RAG lookup          MCP logistics tool
        (keyword stub)        (local docs)          (clinic.json)
```

- **Boundaries:** `services/triage` (API + auth), `agents/` (router + tools), `fixtures/`, `evals/`, `tests/`
- Redaction runs before anything is written to the audit log

4. Skill coverage

Skill	Where in repo	Req/Opt
API / agent path	services/triage	Required
MCP tools	agents/tools/logistics.py	Required
Pin / versioning	prompts/pin.json	Required
Security hygiene	tests/test_security_smoke.py	Required
Eval / test	tests/unit/, evals/golden.jsonl	Required
Docs	docs/	Required
Multimodal	services/catalog	Optional (deferred)

5. Live demo

One command: `bash scripts/demo.sh`

1. `/health` returns healthy
2. Login as coordinator, receive a token
3. Send "child has chest pain" and receive `urgency: high`

Backup: 60-second recording of the same run.

6. Testing evidence

- `pytest -q` : **9 passed, 1 deselected** (integration needs a live service)
- Golden-set eval: **3/3 = 1.00**
- Fixture: **300 rows**, `diagnose OK, all rows agree with router`
- Security smoke: includes a real redaction test (emails, phones, IDs, M-Pesa codes, bearer tokens)
- CI: GitHub Actions runs `pytest`

7. Security and compliance

- JWT login with a coordinator role check (patients get 403)
- Passwords stored as bcrypt hashes; demo credentials are teaching-only
- `.env` and audit logs are kept out of git
- PII redacted before logging
- Synthetic data only, no production scrape

8. Cost table

Call type	Calls (n)	Unit cost	Total	Assumptions
Stub tests (`pytest -q` , CI)	0 live calls	$0	$0	CI never calls a live key
Live demo (`scripts/demo.sh`)	0 live calls	$0	$0	Triage, RAG and the MCP tool all run in stub mode locally
Future live model call (not built)	~10 per demo	teaching estimate, ~$0.10	~$1	Would be capped at about $1; not part of this submission

Teaching table, not a quote. Hosting is the laptop, and GitHub Actions on a public or free-

9. What works vs deferred

Works: stub triage API, auth, router, RAG smoke, one MCP tool, 300-row fixture, eval, CI

Deferred: live model call, catalog service, production performance numbers

Perf figure is a laptop teaching stub, not an SLA.

10. Process

- Feature branch `feature/capstone-setup`, PR to `main`
- Review: self-review with the checklist (no peer pair), logged in `review_log.md`
- **Incident:** fixture row T-016 mislabelled and the router had no "order" keyword. Found with `diagnose_fixture.py`, fixed in config and data
- **Next:** live model behind an env flag, catalog service, whole-word keyword matching